

OKELLO ELLY

Software Developer and Systems Architect specializing in scalable systems, full-stack engineering, and cybersecurity.

CONTACT INFORMATION

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Profile

Software Developer and Systems Architect with hands-on experience building secure, scalable software systems across digital learning, campus services, civic platforms, analytics, and operational workflows. Skilled in full-stack development, backend engineering, REST API design, database architecture, networking, and cybersecurity, with practical implementation of JWT-based authentication and role-based access control (RBAC).

Experienced in designing and delivering high-performance systems with a focus on clean architecture, maintainability, security, and modular scalability. Applied these principles across learning, campus engagement, waste management, and innovation-driven platforms including LearnLight, SmartWaste, Campus Pulse, and ecosystem-based digital solutions.

Proficient in AI-assisted development and data-driven engineering, leveraging modern tooling and analytical thinking to build real-world software systems that solve practical user and infrastructure problems.

Education

Bachelor of Science in Computer Software Engineering

Zetech University

Jan 2025 – Nov 2027

Core Technical Focus

- Software Development (C, C++, JavaScript, Python)
 - Web Development
 - Mobile Application Development
 - Database Design and Management (MySQL, SQLite)
 - Computer Networking
 - System Analysis and Design
 - Software Architecture
 - Software Metrics and Quality Assurance
 - Cybersecurity Fundamentals
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Skills

- Leadership and team collaboration in student programs and software projects
 - Technical communication for documentation, coordination, and team discussions
 - Problem solving and critical thinking in backend development and system design
 - Strategic thinking and initiative in architecture design and project execution
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Work Experience and Projects

BIT System – Backend Developer & System Architect

Technologies: Python (Flask), MySQL, SQLite, JWT, REST APIs, HTML, CSS

Timeline: 2024 – Present

- Designed and developed a campus digital system for student services, event management, payments, and administrative workflows
 - Implemented secure authentication and role-based access control using JWT
 - Built REST APIs supporting structured data access and system operations
 - Designed relational databases to improve data organization, scalability, and query efficiency
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LearnLight – Software Developer

Technologies: React, JavaScript, APIs, Backend Services

Timeline: 2026

- Contributing to a digital learning platform for structured online education and learning access
 - Developing frontend and backend components to support content delivery and user interaction
 - Improving system structure using modular architecture and reusable components
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Campus Pulse – Software Developer

Technologies: TypeScript, PLpSQL, CSS, JavaScript

- Developed a university engagement platform for campus events including hackathons, sports, career fairs, and community activities
 - Built systems to improve event discovery, participation, and student engagement
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SmartWaste – Software Developer

Technologies: Python, Flask, MySQL, REST APIs

- Developed a waste management system for collection tracking and operational reporting
 - Built backend APIs for waste data recording, management, and analytics
 - Designed database structures to support reporting and operational workflows
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SkillBridge Africa (Africa Rise Ecosystem) – Software Developer

Technologies: TypeScript, PLpSQL

- Contributed to a digital ecosystem connecting learning, skills development, mentorship, and real-world opportunities for African students and innovators
 - Built features supporting opportunity discovery, structured learning access, and ecosystem connectivity
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Nairobi Hidden Gems – Software Developer

Technologies: HTML, Web Development

- Developed a Gen Z social discovery platform for aesthetic places and experiences in Nairobi
 - Focused on user-driven discovery and location-based engagement
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My Sim League – Software Developer

Technologies: TypeScript, CSS, JavaScript

- Built a football simulation game offering a realistic managerial experience across top European leagues
 - Designed gameplay systems and progression logic for user engagement
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Certifications

- **Introduction to Modern AI – Cisco Networking Academy (2026)**

AI fundamentals, machine learning concepts, AI applications

- **Junior Cybersecurity Analyst – Cisco Networking Academy**

Threat analysis, vulnerability assessment, SOC fundamentals

- **Huawei Computer Networks (CRA Training Program)**

Network architecture, TCP/IP, routing and switching

- **Cisco Certified Cybersecurity Essentials**

Cyber risk management, secure access control, incident response

- **Data Science and Analytics – HP LIFE (2025)**

Data analysis, visualization, reporting

- **AI Fluency: Framework and Foundations – Anthropic (2025)**

Prompt engineering, AI workflows, AI-assisted development

- **Anticipatory Governance: Systems Thinking and Strategic Foresight Part 1 (2026)**

Systems thinking foundations and governance frameworks

- **Anticipatory Governance: Systems Thinking and Strategic Foresight Part 2 (2026)**

Applied systems thinking and strategic foresight implementation

- **SEO – HubSpot Academy (2025)**

Search optimization and analytics

- **Content Marketing – HubSpot Academy (2024)**

Content strategy and digital storytelling

- **Digital Marketing – HubSpot Academy (2024)**

Campaign strategy and analytics

Volunteer and Leadership

- **Aspire Leaders Program – Cohort 1 Graduate (2026)**

Completed global leadership program focused on systems thinking and professional development

- **Aspire Leaders Program Moderator – Cohort 2 (2026)**

Facilitated discussions and supported participant engagement across cohorts

- **GMI Mentorship Program (2026)**

Participated in structured mentorship focused on leadership and career development

- **Secretary – Innovation Club, Zetech University**

Supported innovation activities, student engagement, and technical initiatives